

Functions

Introduction to computer programming

Management and Artificial Intelligence



Reusing Code

So far...

- we have covered language mechanisms
- we know how to write different `.py` files for different computations
- ...where each file contains some piece of code
- ...and each code is a sequence of operations

PROBLEM: This approach is:

- easy for small-scale problems
- messy for large-scale problems
- hard to keep track of details and maintain
- hard to reuse inside more complex programs

Functions

More code does not necessarily mean good practice → use **functions**.

Why?

- functions are **reusable** chunks of code
- functions run if and only if are “**called**” or “**invoked**”

Characteristics of a function:

- functions have a **name**
- functions have (0 or more) **parameters**
- functions have a **docstring***
- functions have a **body**
- functions return **something**.

*Optional, but recommended

Anatomy of a function

```
def is_even(i):  
    """  
    Input: i, a positive int  
    Returns True if i is even, otherwise False  
    """  
    print("inside is_even")  
    return i%2 == 0  
  
is_even(3)
```

keyword

name

parameters or arguments

specification, docstring

body

later in the code, you call the function using its name and values for parameters

Function body

```
def is_even( i ):
```

```
    """
```

```
    Input: i, a positive int
```

```
    Returns True if i is even, otherwise False
```

```
    """
```

```
    print("inside is_even")
```

```
    return i%2 == 0
```

keyword

*expression to
evaluate and return*

*run some
commands*

When a function is invoked, Python:

- creates a **frame** for the function call
- **assigns** arguments to parameters
- **executes** function body
- **erases** the frame

Example:

```
def f(i):  
    j = i*2  
    return j
```

```
n=f(1)  
n=f(2)  
n=f(3)
```

Let's see it in Python Tutor: pythontutor.com

The importance of `return`

```
def is_even( i ):  
    """  
    Input: i, a positive int  
    Does not return anything  
    """
```

```
i%2 == 0
```

*without a return
statement*

What happens when running e.g. `n=is_even(2)` ?

- Python returns `None` if no `return` is given
- `None` represents the absence of a value

print vs. return

print

- can be placed **anywhere**
- as **many** `print` statements as you want
- code **after** `print` is executed
- has a value **outputted** to the console

return

- has a meaning only **inside** a function
- only **one** `return` statement is executed
- code **after** `return` is not executed
- has a value **returned** to the function caller

One return, multiple values

The `return` statement can return multiple values (if needed):

- list of values separated by commas
- returns a tuple*

Example:

```
def hms(nsec):  
    """Takes nsec (n. of seconds) and splits in hours, minutes, seconds"""  
    hh = nsec // 3600  
    nsec = nsec % 3600  
    mm = nsec // 60  
    ss = nsec % 60  
    return hh, mm, ss  
  
print(hms(4000))  
print(hms(100000))
```

```
(1, 6, 40)  
(27, 46, 40)
```

*We will see tuples later in the course, for now we just print multiple outputs.

Variable Scope

```
def f(x):  
    x = x + 1  
    print('in f(x): x =', x)  
    return x
```

*formal
parameter*

*Function
definition*

```
x = 3  
z = f(x)
```

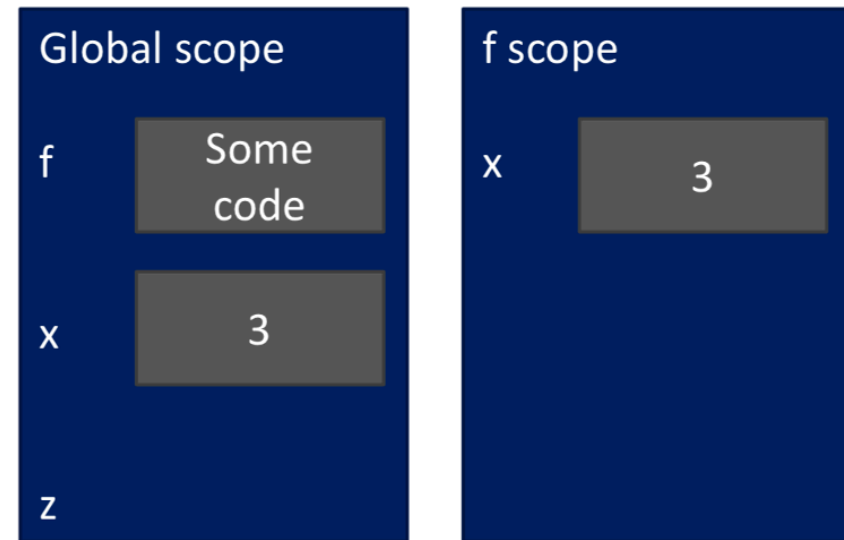
*actual
parameter*

Main program code
** initializes a variable x*
** makes a function call f(x)*
** assigns return of function to variable z*

- the **formal parameter** gets bound to the **actual parameter** when the function is called
- when entering a function, a new **scope** (or **frame**, or **environment**) is created
- the **scope** is the mapping of names to objects

Running Example (1/4)

```
def f( x ) :  
    x = x + 1  
    print('in f(x): x =', x)  
    return x  
  
x = 3  
z = f( x )
```



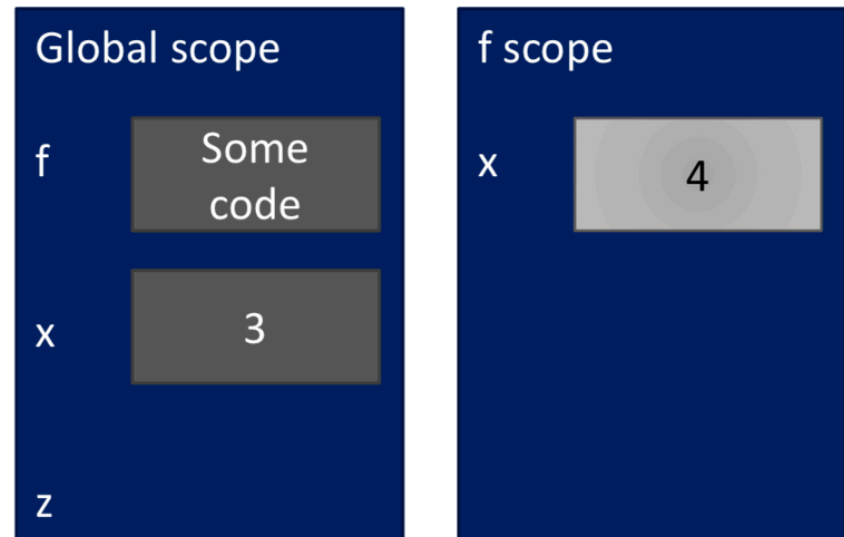
What's going on here?

1. in the main (i.e., global) scope, we define function `f`
2. in the main scope, we define variable `x`
3. function `f` is called and variable `x` is fed to `f`

Running Example (2/4)

```
def f( x ):  
    x = x + 1  
    print('in f(x): x =', x)  
    return x
```

```
x = 3  
z = f( x )
```



What's going on here?

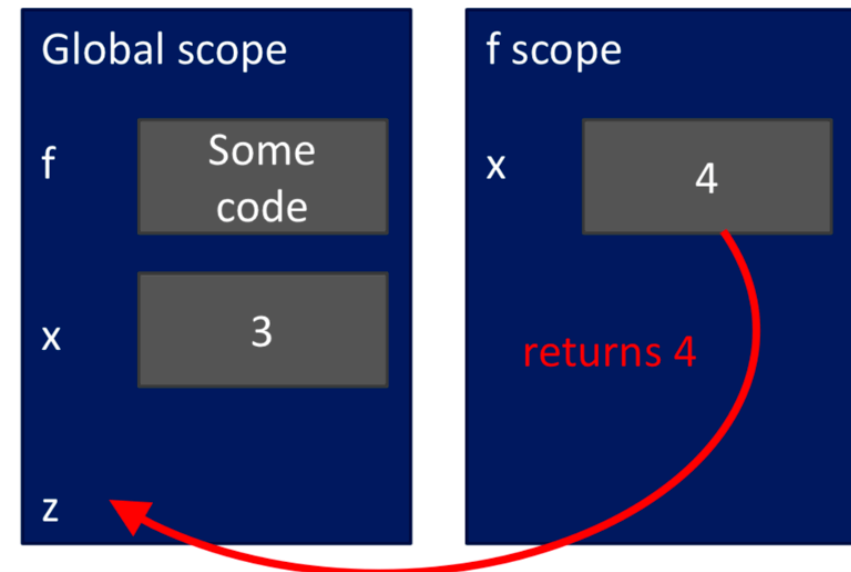
1. variable `x` is changed inside `f` because of `x = x+1`

NOTE: Variable `x` in the global scope has not been changed because `f` has its own scope!

Running Example (3/4)

```
def f( x ) :  
    x = x + 1  
    print('in f(x): x =', x)  
    return x
```

```
x = 3  
z = f( x )
```



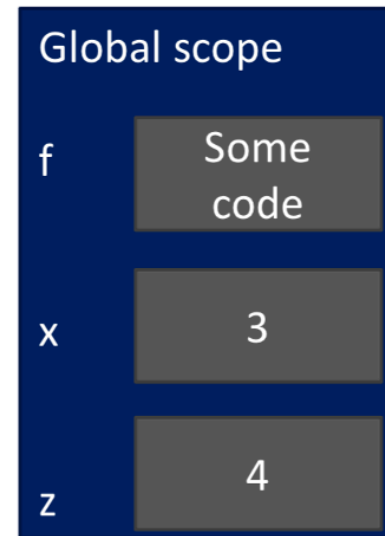
What's going on here?

1. variable `x` inside `f` is ready to be returned
2. in the main scope, there is variable `z` ready to host the returned value

Running Example (4/4)

```
def f( x ) :  
    x = x + 1  
    print('in f(x): x =', x)  
    return x
```

```
x = 3  
z = f( x )
```



What's going on here?

1. the frame of function `f` is destroyed because its execution is finished
2. in the main scope, we have the 'original' value for `x`
3. in the main scope, we have the output of function `f` in the new variable `z`

Important Caveats on Variable Scopes (part 1)

```
def f(y):  
    x = 1  
    x += 1  
    print(x)  
  
x = 5  
f(x)  
print(x)
```

*x is re-defined
in scope of f*

*different x
objects*

```
def g(y):  
    print(x)  
    print(x + 1)  
  
x = 5  
g(x)  
print(x)
```

*x from
outside g*

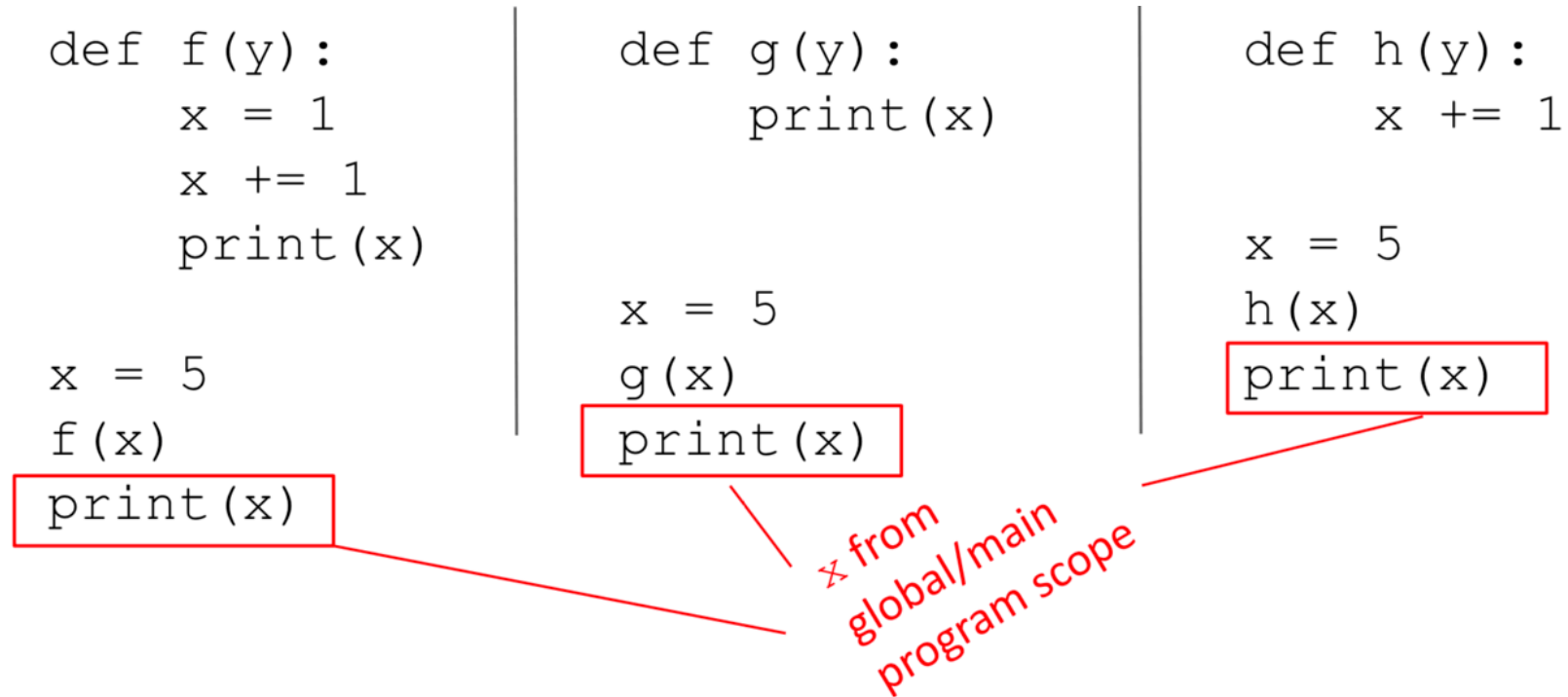
*x inside g is picked up
from scope that called
function g*

```
def h(y):  
    x += 1  
  
x = 5  
h(x)  
print(x)
```

*UnboundLocalError: local variable
'x' referenced before assignment*

- inside a function, you **can access** variables defined outside the function itself
- inside a function, you **cannot modify** variables defined outside the function itself
- to modify variables defined outside, you must use the `global` keyword... which is frowned upon

Important Caveats on Variable Scopes (part 2)



- due to the fact that (by default) you cannot modify variables inside a function, `x` remains untouched

The `global` Keyword

- when we create a variable `inside` a function, it is `local` by default
- when we create a variable `outside` a function, it is `global` by default
- using the `global` keyword outside a function has no effect
- we use the `global` keyword to *write/modify* a variable `inside` a function

For instance, let us consider this code:

```
def add():  
    global c  
    c = 3  
    print("Inside add(): c = ", c)  
  
c = 1  
add()  
print("In main(): c = ", c)
```

Inside add(): c = 3

In main(): c = 3

Function `add` is modifying the global variable `c`.

What happens if we omit the `global` keyword?

```
def add():  
    c = 3 # this a local variable!  
    print("Inside add(): c = ", c)  
  
c = 1 # this is a gloabl variable  
add()  
print("In main(): c = ", c)
```

```
Inside add(): c = 3  
In main(): c = 1
```

Function `add` is not modifying the global variable `c` but its own local variable.

Nested Functions

- You can **nest** functions arbitrarily.
- Scopes of nested functions behave like Matryoshka dolls.

Let's see how thanks to pythontutor.com on this example:

```
def g(x):  
    def h():  
        x = 'abc'  
    x = x + 1  
    print('g: x =', x)  
    h()  
    return x  
  
x = 3  
z = g(x)
```

g: x = 4

Positional vs. Keyword Arguments

We have seen **positional bounding** of formal parameters to actual parameters:

- the first formal parameter is bound to the first actual parameter
- the second formal to the second actual
- ...and so on, until all parameters are bounded

Python also supports **keyword bounding**

- formals are bound to actuals using the name of the formal parameter

Example:

```
def add(a, b, c):  
    return a + b + c  
  
print(add(1, 2, 3))           # positional bounding a->1, b->2, c->3  
print(add(c=3, a=1, b=2))    # keyword bounding (order does not matter)
```

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6

Optional and Default Parameters

All parameters need to be provided to a function:

- via positional or keyword arguments
- via default parameters

For instance, when we consider this function:

```
def area_cylinder(radius=1, height=1):  
    pi = 3.14  
    bottom_area = pi * radius ** 2  
    side_area = pi * height * radius * 2  
    return side_area + bottom_area * 2
```

When we call this function with no parameters:

```
def area_cylinder(radius=1, height=1):  
    pi = 3.14  
    bottom_area = pi * radius ** 2  
    side_area = pi * height * radius * 2  
    return side_area + bottom_area * 2  
  
print("No parameters: ", area_cylinder())
```

No parameters: 12.56

The function will use the default values for the two parameters. Indeed:

```
print("Check: ", area_cylinder(1, 1))
```

Check: 12.56

Or, we can call this function with one parameter:

```
def area_cylinder(radius=1, height=1):  
    pi = 3.14  
    bottom_area = pi * radius ** 2  
    side_area = pi * height * radius * 2  
    return side_area + bottom_area * 2  
  
print("One parameter: ", area_cylinder(2))
```

One parameter: 37.68

We are passing only the first (positional) parameter. For the second one we are using the default value. Indeed:

```
print("Check: ", area_cylinder(2, 1))
```

Check: 37.68

As already seen, the natural choice would be to call it with the two parameters:

```
def area_cylinder(radius=1, height=1):  
    pi = 3.14  
    bottom_area = pi * radius ** 2  
    side_area = pi * height * radius * 2  
    return side_area + bottom_area * 2  
  
print("Two parameters: ", area_cylinder(2, 3))
```

Two parameters: 62.8

If we want to set only one specific parameter (e.g., the second one), while keeping the default values for the other ones (e.g., the first parameter), then we can:

```
def area_cylinder(radius=1, height=1):  
    pi = 3.14  
    bottom_area = pi * radius ** 2  
    side_area = pi * height * radius * 2  
    return side_area + bottom_area * 2  
  
print("Specific parameters: ", area_cylinder(height=2))
```

Specific parameters: 18.84

Which would be equivalent to:

```
print("Check: ", area_cylinder(1, 2))
```

Check: 18.84

Built-in Functions

Python implements many functions by default, which are called **built-in functions**.

Examples of built-in functions:

- numerical functions, such as `round(x)` and `pow(x,y)` *
- casting, such as `int(x)`, `float(x)` and `str(x)`
- input and output, namely `input()` and `print()`
- help function, namely `help()` **

Related functions are grouped into files, called **modules**.

* Which is another way of doing exponentiation: `pow(x,y)` is the same as `x**y`. ** The `help()` function takes as input the name of a function and returns its docstring.

Modules

Modular programming is the way of breaking a large, unwieldy programming task into separate, smaller (and more manageable) subtasks.

A **module** is a piece of software with a specific functionality. For example, a game (e.g., chess, ping-pong, ...):

- a module responsible for the game logic (rules)
- a module responsible for graphics

In Python, each module is a Python file with a `.py` extension, yet other extensions are also possible (e.g., pre-compiled C++).

Modules contain some Python code (e.g., functions, class definitions, ...).

For instance, a `random` module might look like this:

File: `random.py`

```
def sample(args):  
    """Chooses k unique random elements from a population..."""  
    # some code  
  
def randint(args):  
    """Return random integer in range [a, b], including both end points."""  
    # some code  
  
def random(args):  
    """Get the next random number in the range [0.0, 1.0)."""  
    # some code
```

Using Modules in Python

Modules are **imported** (in your script, in your function or in other modules) using the `import` command.

You can use two different strategies for using a specific function from that module.

1. using the `<module>.<function>` notation

```
import random                # or any other module
print(random.randint(1, 100)) # or any other function
```

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2. using the `from` command

```
from random import randint # or any other function
print(randint(1, 100))
```

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WARNING: Strategy #2 does not need the dot-notation (see Strategy #1). Yet, it only loads a specific function in your main!

Common Modules in Python

Python comes with many modules, including:

- `io` : read and write from files
- `math` : mathematical functions
- `random` : pseudo-random number generator
- `string` : useful string functions
- `sys` : information about your operating system

You can find the full list here:

<https://docs.python.org/3/library/>

An example: the `math` module:

- Number-theoretic and representation functions
- Power and logarithmic functions
- Trigonometric functions
- Angular conversion
- Hyperbolic functions
- Special functions and constants

```
>>> import math
>>> print(math.factorial(x))      # return x!
>>> print(math.log2(x))          # return the base-2 logarithm of x
>>> print(math.cos(math.pi))     # return the cosine of pi
>>> print(math.gcd(x,y))         # return the greatest common divisor
```

You can find the full list here:

<https://docs.python.org/3/library/math.html>

Interactive Documentation

You can browse the contents of a module interactively via the `dir()` function.

```
import math
print(dir(math))
```

```
['__doc__', '__file__', '__loader__', '__name__', '__package__', '__spec__', 'acos', 'acosh', 'asin', 'asinh', 'atan', 'atan2', 'atanh', 'cbrt', 'ceil', 'comb', 'copysign', 'cos', 'cosh', 'degrees', 'dist', 'e', 'erf', 'erfc', 'exp', 'exp2', 'expm1', 'fabs', 'factorial', 'floor', 'fmod', 'frexp', 'fsum', 'gamma', 'gcd', 'hypot', 'inf', 'isclose', 'isfinite', 'isinf', 'isnan', 'isqrt', 'lcm', 'ldexp', 'lgamma', 'log', 'log10', 'log1p', 'log2', 'modf', 'nan', 'nextafter', 'perm', 'pi', 'pow', 'prod', 'radians', 'remainder', 'sin', 'sinh', 'sqrt', 'sumprod', 'tan', 'tanh', 'tau', 'trunc', 'ulp']
```

NOTE: Items beginning/ending with `__` are some metadata files. You can safely ignore them. For example, `__file__` contains the path pointing to the module source file.

Given a list of functions, you can browse their documentation via the `help()` function.

RECALL: The `help()` function takes as input the name of a function and returns its docstring.

```
>>> import math
>>> help(math.log)
Help on built-in function log in module math:

log(...)
    log(x, [base=math.e])
    If the base not specified, returns the natural logarithm (base e) of x.

(END)
```

